

Senri Nakamura

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SKILLS

Languages: IEC 6113-3(LAD, FBD, SCL, STL), C, C++, Python, VHDL, Verilog

Software: Siemens TIA Portal, Siemens Industrial Edge Hub, Node-RED, Docker, CODESYS, Grafana, SCADA, HMI

Protocol: Modbus RTU/TCP, RS-232/422/485, CAN, Ethernet/IP, Profibus, MQTT, OPC UA, NTP sync, UART, I²C

EDUCATION

Tufts University

Medford, MA

B.S. Electrical Engineering, GPA:3.5/4.0

Sept 2022 - May 2026

Coursework: Embedded Systems, Digital Logic, Electronics I & II, Machine Learning, CMOS design, EM Fields & Waves

WORK EXPERIENCE

Interstellar Technologies Inc.

Tokyo, JPN

Automation Engineer Intern

June 2025 - Aug 2025

- Programmed Siemens ET200SP PLCs (TIA Portal v18/STEP7), implementing Modbus RTU/TCP and P2P.
- Developed CRC32-ISO-HDLC validation logic and NTP-based time sync.
- Configured SCADA/HMI systems and Grafana dashboards, enabling real-time monitoring.

Avionics Engineer Intern

June 2025- Aug 2025

- Developed FPGA IP on Xilinx Zynq-7000 with UART/RS-232/422/485 interfaces for 10+ subsystems.
- Configured multi-board interface with M12/M16 connectors for 2 engine integration tests.

Eco-Pork Inc.

Tokyo, JPN

Data (Contract)

May 2025 - July 2025

- Validated structured and unstructured data to ensure reliability in automation, ML, and AI workflows.

Recruit Holdings Co., Ltd.

Tokyo, JPN

Product Manager Intern

July 2024 - Aug 2024

- Led end-to-end product development with 100+ customer interviews in a cross-functional team of 3.
- Built an HR platform for the 50+ market using agile methods, targeting a \$1B TAM/\$100M SOM.
- Delivered MVP for full-time development, advancing it to the experiential phase with executive buy-in.

Nano Lab - Tufts University

Medford, MA

Undergraduate Research Assistant

Apr 2023 - Sep 2023

- Examined triboelectric nanogenerators and pressure sensors in the gastrointestinal (GI) tract via an ingestible pill.

PROJECT EXPERIENCE

PLC Log Intelligence Platform (Siemens Kicks for Edge Hackathon - Global Finalist)

- Built an analytics system on SIMATIC IPC BX-39A using Node-RED and OpenSearch to ingest, decompress, and parse 1M+ PLC/MFC logs into structured JSON and real-time dashboards, reducing analysis time from 7+ days to less than 10 minutes.

Tufts Solar Vehicle Project (TSVP)

- Led production of 33 solar arrays (256 cells) with 95% alignment and 75% power performance.
- Reduced costs by 99.95% by building an in-house EL-test setup and lamination & pre-shrinking machine.

Ultra-Low-Power CMOS OTA for Capacitive-Touch Sensing in Wearable Interfaces

- Designed in XFAB 180nm achieving 57 dB gain, ~20 MHz GBW, and 180μW.

Pulse Oximeter with Feedback Control

- Developed a pulse oximeter with MOSFET LED control and band-pass filters for accurate SpO₂ readings.

LEADERSHIP EXPERIENCE

Tufts Solar Vehicle Project (TSVP)

Medford, MA

President

Aug 2024 - Aug 2025

- Grew the team by 1000% (120 members), created the first managerial system for 3 subteams, 15+ subgroups, and 12 leads.
- Managed \$200K budget, raised \$100K additional funding, 45+ sponsors, and secured 3 dedicated workspaces (first of 400+ orgs).

Electrical Co-Lead, Founding Member (1 of 6)

Sep 2023 - Jul 2024

- Supervised a 4-person team, coordinating cross-team integration to align design and system testing.